

Alcohols, Phenols and Ethers

NCERT Class XII Chemistry · Chapter 7 · Alcohols, Phenols and Ethers

Alcohols, Phenols and Ethers — Complete Notes

6 marks in the board paper (CBSE 2026-27 curriculum). Three families of compounds, all built round one oxygen atom: an alcohol has an $-OH$ on a carbon chain, a phenol has an $-OH$ on a benzene ring, an ether has the oxygen sitting between two carbon groups. Most of the chapter is one idea used again and again — what the oxygen does with its lone pairs, and which of its two bonds ($O-H$ or $C-O$) breaks. Every new word is tagged [exposure] and explained from scratch. The reactions come in a fixed list; learn the reagent that goes with each and the marks follow.

What the three families are, and where they matter — *one oxygen atom, three different places to put it*

[exposure] A **hydrocarbon** is a compound of only carbon and hydrogen (methane CH_4 , benzene C_6H_6). Replacing a hydrogen by another atom or group gives a new compound with new properties. The group that carries the property is called the **functional group**: it is the part of the molecule where the chemistry happens. In this chapter the functional group is built on oxygen.

[exposure] The **hydroxyl group** is $-OH$: an oxygen bonded to a hydrogen, and to the rest of the molecule through the oxygen.

- **Alcohol** — an $-OH$ on a carbon of an **aliphatic** system (a chain or a non-aromatic ring of carbons). Example: methanol CH_3OH , ethanol C_2H_5OH .
- **Phenol** — an $-OH$ on a carbon of an **aromatic** system (a benzene ring). Example: phenol itself, C_6H_5OH .
- **Ether** — the hydrogen of the $-OH$ is replaced by an alkyl or aryl group, so the oxygen links two carbon groups: $R-O-R'$. Example: methoxymethane CH_3OCH_3 . You can picture an ether as an alcohol or phenol whose hydroxyl hydrogen has been swapped for a carbon group.

Why the chapter matters outside the exam: alcohols, phenols and ethers are the starting points for **detergents**, **antiseptics** and **fragrances** respectively. Ethanol is the spirit used to polish wood; the sugar in food, the cotton in cloth and the paper in books all contain many $-OH$ groups.

Examiner asks: nothing directly from this opening, but "alcohol or phenol?" depends on whether the carbon holding the $-OH$ is in a ring of benzene type or not, and that decides almost every property later.

Classification — *count the $-OH$ groups; then look at the carbon that carries them*

[exposure] **Monohydric, dihydric, trihydric, polyhydric** only say how many $-OH$ groups a molecule has: one, two, three, or many. This applies to alcohols and to phenols.

Number of –OH	Alcohol example	Phenol example
Monohydric (1)	ethanol $\text{CH}_3\text{CH}_2\text{OH}$	phenol $\text{C}_6\text{H}_5\text{OH}$
Dihydric (2)	ethane-1,2-diol $\text{HOCH}_2\text{CH}_2\text{OH}$	catechol, resorcinol, hydroquinone
Trihydric (3)	glycerol (propane-1,2,3-triol)	—

Classifying a monohydric alcohol by the carbon that carries –OH

[exposure] Hybridisation is the way a carbon atom mixes its orbitals before it bonds. A carbon with four single bonds is sp^3 hybridised; a carbon in a benzene ring or a $\text{C}=\text{C}$ (three attached groups, one double bond) is sp^2 hybridised. The first rule: ask whether the –OH carbon is sp^3 or sp^2 .

The –OH is on an sp^3 carbon:

- **[exposure] Primary (1°)** alcohol: the –OH carbon is attached to **one** other carbon (or none, as in CH_3OH). **Secondary (2°)**: attached to **two** carbons. **Tertiary (3°)**: attached to **three** carbons. Examples: $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ (1°), $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$ (2°), $(\text{CH}_3)_3\text{C}-\text{OH}$ (3°). This count decides how an alcohol oxidises, dehydrates and reacts with HCl , so you will use it all chapter.
- **[exposure] Allylic alcohol**: the sp^3 carbon bearing –OH is directly next to a $\text{C}=\text{C}$ double bond, as in $\text{CH}_2=\text{CH}-\text{CH}_2\text{OH}$. **Benzylic alcohol**: that sp^3 carbon is directly next to a benzene ring, as in $\text{C}_6\text{H}_5-\text{CH}_2\text{OH}$. Allylic and benzylic alcohols can each be primary, secondary or tertiary.

The –OH is on an sp^2 carbon: the –OH sits on a carbon of a $\text{C}=\text{C}$, either an ordinary one (a **[exposure] vinylic alcohol**, $\text{CH}_2=\text{CH}-\text{OH}$) or a benzene-ring carbon (a phenol). These are a different class and behave very differently, as the acidity section shows.

Classifying ethers

[exposure] If the two groups on the oxygen are the same the ether is **simple (symmetrical)**: $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$, diethyl ether. If they differ it is **mixed (unsymmetrical)**: $\text{C}_2\text{H}_5\text{OCH}_3$ and $\text{C}_2\text{H}_5\text{OC}_6\text{H}_5$. This matters because the two kinds are made and cleaved differently.

NCERT intext 7.1 and 7.2 ask you to classify six drawn alcohols. NCERT's answers: primary (i), (ii), (iii); secondary (iv), (v); tertiary (vi); allylic (ii) and (vi). The method is always the same: find the –OH carbon, count the carbons attached to it, then check for a neighbouring $\text{C}=\text{C}$.

Examiner asks: "classify as primary, secondary or tertiary", and later "which of these is a tertiary alcohol?" in a reasoning question. Count the carbon neighbours of the –OH carbon.

Nomenclature — *common names are quick; IUPAC names carry the marks*

Alcohols

[exposure] IUPAC is the international system for naming compounds, so every compound has exactly one correct name. For an alcohol: take the longest carbon chain that **contains the carbon bearing –OH**, name the matching alkane, **drop its final "e" and add "ol"**. Number from the end **nearest to the –OH**, and write the position of –OH and of every other group.

- Common name: name the alkyl group, then the word **alcohol**: CH_3OH methyl alcohol; $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ n-propyl alcohol.
- Two or more –OH groups: **keep the "e"** of the alkane and add **diol, triol** with locants: $\text{HOCH}_2\text{CH}_2\text{OH}$ = **ethane-1,2-diol** (ethylene glycol). Glycerol is **propane-1,2,3-triol**.

- Cyclic alcohols: prefix **cyclo**, with the carbon bearing –OH as C-1: cyclohexanol; 2-methylcyclopentanol.

Compound	Common name	IUPAC name
CH ₃ OH	methyl alcohol	methanol
CH ₃ CH ₂ CH ₂ OH	n-propyl alcohol	propan-1-ol
CH ₃ CH(OH)CH ₃	isopropyl alcohol	propan-2-ol
CH ₃ CH ₂ CH ₂ CH ₂ OH	n-butyl alcohol	butan-1-ol
CH ₃ CH ₂ CH(OH)CH ₃	sec-butyl alcohol	butan-2-ol
(CH ₃) ₂ CHCH ₂ OH	isobutyl alcohol	2-methylpropan-1-ol
(CH ₃) ₃ C–OH	tert-butyl alcohol	2-methylpropan-2-ol
HOCH ₂ CH ₂ OH	ethylene glycol	ethane-1,2-diol

Worked example · NCERT Example 7.1(i)

Name CH₃–CH(Cl)–CH(CH₃)–CH(CH₃)–CH₂OH .

1. Longest chain containing the –OH carbon: five carbons, so **pentane** → **pentan-1-ol** (the CH₂OH end is C-1, the end nearest –OH).
2. Substituents: CH₃ on C-2 and C-3, Cl on C-4.
3. Write them alphabetically with the parent: chloro before methyl.

Answer: **4-chloro-2,3-dimethylpentan-1-ol**.

Phenols

The simplest hydroxy derivative of benzene is **phenol**; that is its common name and an accepted IUPAC name. In substituted phenols the ring carbon bearing –OH is C-1, and the positions are called **ortho** (neighbour, 1,2), **meta** (1,3) and **para** (opposite, 1,4). Methylphenols are **cresols**: o-cresol = 2-methylphenol; m-cresol = 3-methylphenol; p-cresol = 4-methylphenol. The dihydroxybenzenes are **catechol** (benzene-1,2-diol), **resorcinol** (benzene-1,3-diol) and **hydroquinone** or quinol (benzene-1,4-diol). Example: 2,6-dimethylphenol.

Ethers

- **Common name:** the two groups in alphabetical order as separate words, then **ether**: CH₃OC₂H₅ ethyl methyl ether. Identical groups: **di-**: C₂H₅OC₂H₅ diethyl ether.
- **IUPAC name:** treat the ether as a hydrocarbon carrying an **alkoxy** group –OR (or aryloxy –OAr). The **larger group** is the parent hydrocarbon; the smaller one becomes "methoxy", "ethoxy" and so on.

Compound	Common name	IUPAC name
CH ₃ OCH ₃	dimethyl ether	methoxymethane
C ₂ H ₅ OC ₂ H ₅	diethyl ether	ethoxyethane
CH ₃ OCH ₂ CH ₂ CH ₃	methyl n-propyl ether	1-methoxypropane
C ₆ H ₅ OCH ₃	methyl phenyl ether (anisole)	methoxybenzene
C ₆ H ₅ OC ₂ H ₅	ethyl phenyl ether (phenetole)	ethoxybenzene
CH ₃ –CH(CH ₃)–O–CH ₃	methyl isopropyl ether	2-methoxypropane
CH ₃ OCH ₂ CH ₂ OCH ₃	—	1,2-dimethoxyethane

Other NCERT answers worth knowing: $\text{CH}_3-\text{CH}(\text{CH}_3)-\text{O}-\text{CH}_2\text{CH}_3$ is **2-ethoxypropane** (Example 7.1(ii)); the ring compound with OC_2H_5 and NO_2 on neighbouring carbons is **1-ethoxy-2-nitrocyclohexane**; intertext 7.3 answers are 4-chloro-3-ethyl-2-(1-methylethyl)butan-1-ol, 2,5-dimethylhexane-1,3-diol, 3-bromocyclohexanol, hex-1-en-3-ol and 2-bromo-3-methylbut-2-en-1-ol.

Trap: when an alcohol also has a double bond, the $-\text{OH}$ gets the lowest number, even if the double bond then gets a high one. $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{CH}=\text{CH}_2$ is **hex-5-en-2-ol** (numbered from the $-\text{OH}$ end), not hex-1-en-5-ol. Both "en" and "ol" appear in the name, with the "e" of "ene" dropped before the vowel of "ol".

Examiner asks: "write the IUPAC name of the following" — a standing 2-marker (for example 3-phenylprop-2-en-1-ol, 2017; 2-methoxy-2-methylpropane, 2019). See the PYQ section.

Structure of the functional groups — *why the bond angles and bond lengths come out as they do*

[exposure] A **sigma (σ) bond** is a single bond formed by head-on overlap of two orbitals. In an alcohol the oxygen of $-\text{OH}$ is sp^3 hybridised and joins the sp^3 carbon through a sigma bond. **[exposure]** A **lone pair** is a pair of valence electrons on an atom that is not used in bonding. Oxygen has two lone pairs, and they are what make alcohols, phenols and ethers react.

NCERT Fig. 7.1 gives the numbers to remember:

Compound	Bond angle	C–O bond length
Methanol CH_3OH	C–O–H = 108.9°	142 pm
Phenol $\text{C}_6\text{H}_5\text{OH}$	C–O–H = 109°	136 pm
Methoxymethane CH_3OCH_3	C–O–C = 111.7°	141 pm

- **Alcohols: angle slightly below the tetrahedral 109.5° (109°28').** The two lone pairs on oxygen repel the bonded pairs and squeeze the C–O–H angle. Lone pair against bond is a stronger repulsion than bond against bond.
- **Phenol: C–O is shorter (136 pm against 142 pm in methanol).** Two reasons: (i) the oxygen's lone pair is shared with the benzene ring (**[exposure] conjugation:** a lone pair next to a ring spreads onto it), which gives the C–O bond **partial double-bond character**; (ii) the carbon is sp^2 hybridised, and an sp^2 carbon holds its bonds a little closer than an sp^3 carbon.
- **Ethers: angle slightly above tetrahedral (111.7°).** The two bulky alkyl groups on oxygen repel each other and push the angle open. The C–O length (141 pm) is about the same as in alcohols.

Examiner asks: two reasons that come up directly: "why is the C–O bond in phenol shorter than in methanol?" (2023) and "why is the C–O–C angle in ethers slightly greater than the tetrahedral angle?" (2025). Write the reason, not just the numbers.

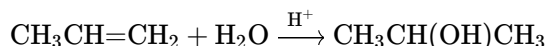
Preparing alcohols from alkenes — *two ways to add water across a C=C — one follows Markovnikov, one does the opposite*

[exposure] An **alkene** has a $\text{C}=\text{C}$ double bond. **Hydration** means adding water ($\text{H}-\text{OH}$) across it, so one carbon takes an H and the other takes an OH; the product is an alcohol. **[exposure] Markovnikov's rule** says that when a reagent $\text{H}-\text{X}$ adds to an **unsymmetrical** alkene (the two carbons of $\text{C}=\text{C}$ carry different

groups), the H goes to the carbon that already has **more** hydrogens and the negative part goes to the carbon with **fewer** hydrogens. You met it in haloalkanes.

1. Acid-catalysed hydration (Markovnikov)

An alkene and water in the presence of acid (H^+) as a **[exposure] catalyst** (a substance that speeds up a reaction and is given back unchanged at the end).



Here OH goes to the carbon with fewer hydrogens, giving **propan-2-ol**.

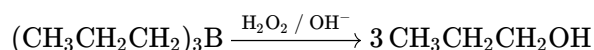
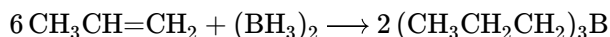
Mechanism, three steps (keep going until the stable product forms):

1. **Protonation of the alkene to a carbocation, by electrophilic attack of H_3O^+** . First the acid makes the hydronium ion: $\text{H}_2\text{O} + \text{H}^+ \longrightarrow \text{H}_3\text{O}^+$. **[exposure]** An **electrophile** is a species that is short of electrons and attacks electron-rich places; here the H^+ attacks the π electrons of the double bond. A **[exposure] carbocation** is a carbon that has lost a bonding pair and carries a positive charge, with only three bonds. $\text{CH}_2=\text{CH}_2 + \text{H}_3\text{O}^+ \longrightarrow \text{CH}_3-\text{CH}_2^+ + \text{H}_2\text{O}$.
2. **Nucleophilic attack of water on the carbocation**. A **[exposure] nucleophile** is an electron-rich species with a lone pair that attacks positive carbon. The oxygen of a second water molecule uses its lone pair to bond to the positive carbon: $\text{CH}_3\text{CH}_2^+ + \text{H}_2\text{O} \longrightarrow \text{CH}_3\text{CH}_2-\text{OH}_2^+$. The oxygen now carries a positive charge.
3. **Deprotonation to the alcohol**. A third water molecule takes the H^+ from that oxygen, giving $\text{CH}_3\text{CH}_2\text{OH}$ and regenerating H_3O^+ .

The acid is given back at the end, which is why it counts as a catalyst. A tertiary carbocation is easier to form than a secondary one, so the rule "hydrogen to the carbon with more hydrogens" is really a rule about forming the more stable carbocation.

2. Hydroboration–oxidation (anti-Markovnikov)

[exposure] Diborane is $(\text{BH}_3)_2$, behaving like two molecules of BH_3 . It adds to an alkene to give a **trialkylborane** R_3B (three alkene molecules add to one boron). Treating this with hydrogen peroxide and aqueous sodium hydroxide **oxidises** it to an alcohol.



The boron attaches to the sp^2 carbon that carries **more hydrogens**, so the alcohol looks as if water had added **opposite to Markovnikov's rule**: propene gives **propan-1-ol** here but propan-2-ol by acid hydration. The yield is excellent. The method was reported by H. C. Brown in 1959; he shared the 1979 Nobel Prize in Chemistry for it.

Trap: "propene to propan-1-ol" needs **hydroboration–oxidation** (B_2H_6 , then $\text{H}_2\text{O}_2/\text{OH}^-$); "propene to propan-2-ol" is **acid-catalysed hydration** (H_3O^+). A question that names only the product decides the reagent for you — check which carbon the OH is on.

Examiner asks: the mechanism of hydration of ethene (2017, asked twice), and the product of hydroboration–oxidation of propene (2020). The mechanism is the three steps above; write each step's name, then the species.

Preparing alcohols from carbonyl compounds and Grignard reagents – *reduction goes down from C=O; Grignard builds the carbon chain*

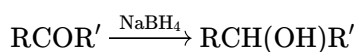
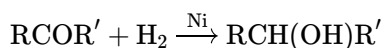
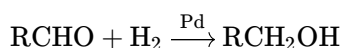
[exposure] A **carbonyl compound** contains the **carbonyl group** C=O: an **aldehyde** RCHO, a **ketone** RCOR', a **carboxylic acid** RCOOH, or an **ester** RCOOR'. These are the next chapter's topic, so here you only need what they become. **[exposure] Reduction** of an organic compound means adding hydrogen to it (or removing oxygen).

1. Reducing aldehydes and ketones

Two kinds of reducing method:

- **Catalytic hydrogenation:** add H₂ over a finely divided metal catalyst (platinum, palladium or nickel).
- **Chemical reducing agents:** sodium borohydride NaBH₄ or lithium aluminium hydride LiAlH₄.

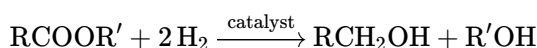
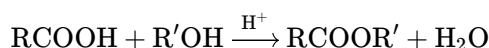
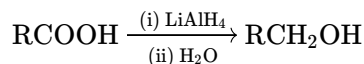
An **aldehyde** gives a **primary** alcohol; a **ketone** gives a **secondary** alcohol.



[exposure] NaBH₄ is a **mild reducing agent**: it reduces aldehydes and ketones but **not** carboxylic acids, esters or amides. LiAlH₄ is a strong reducing agent and reduces all of them. This difference is a popular exam trap.

2. Reducing carboxylic acids and esters

Acids are reduced to primary alcohols, in excellent yield, by LiAlH₄ (then water). LiAlH₄ is **expensive**, so it is used only for special chemicals. To make an alcohol in bulk, first turn the acid into an **ester** (reacting it with an alcohol under acid), then reduce the ester by catalytic hydrogenation. That also hands back the alcohol you used.



Example: ethanoic acid with ethanol gives ethyl ethanoate; hydrogenation over platinum gives **two molecules of ethanol**, one of which was the alcohol you started with.

3. Grignard reagents with aldehydes and ketones

[exposure] A **Grignard reagent** is RMgX ($\text{X} = \text{Cl}, \text{Br}, \text{I}$): an alkyl or aryl group bonded to magnesium. Magnesium is a metal (electropositive), so the carbon on it carries a partial negative charge (δ^-) and behaves as a **nucleophile**. The $\text{C}=\text{O}$ carbon carries a partial positive charge (δ^+) because oxygen is more electronegative and pulls the electrons. So the Grignard carbon attacks the carbonyl carbon.

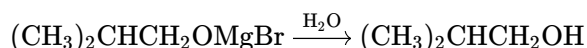
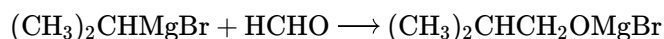
1. **Nucleophilic addition** gives an **[exposure] adduct** (the addition product): the π electrons of $\text{C}=\text{O}$ move onto oxygen, which takes MgX and becomes $-\text{O}-\text{MgX}$.
2. **Hydrolysis** (adding water or dilute acid) replaces MgX by H and gives the alcohol, with $\text{Mg}(\text{OH})\text{X}$ as by-product.

Carbonyl compound	Alcohol class	Equation
Methanal HCHO	primary	$\text{HCHO} + \text{RMgX} \longrightarrow \text{RCH}_2\text{OMgX} \xrightarrow{\text{H}_2\text{O}} \text{RCH}_2\text{OH}$
Any other aldehyde	secondary	$\text{R}'\text{CHO} + \text{RMgX} \longrightarrow \text{R}'\text{CH}(\text{R})\text{OMgX} \xrightarrow{\text{H}_2\text{O}} \text{R}'\text{CH}(\text{OH})\text{R}$
A ketone	tertiary	$\text{R}'\text{COR}'' + \text{RMgX} \longrightarrow \text{R}'\text{R}''\text{C}(\text{R})\text{OMgX} \xrightarrow{\text{H}_2\text{O}} \text{R}'\text{R}''\text{C}(\text{R})\text{OH}$

Remember this table; it is a standing question. It also tells you how to **design** a Grignard route: the R on the Grignard reagent is the group that ends up on the carbon that carries the $-\text{OH}$.

Worked example · NCERT intext 7.4 — make 2-methylpropan-1-ol and cyclohexylmethanol from methanal

1. Both alcohols are primary and end in $-\text{CH}_2\text{OH}$, so the CH_2OH comes from **methanal**.
2. The rest of the molecule is the group that sat on magnesium. For $(\text{CH}_3)_2\text{CH}-\text{CH}_2\text{OH}$ the group is isopropyl: Grignard reagent $(\text{CH}_3)_2\text{CH}-\text{MgBr}$.
3. For cyclohexylmethanol it is cyclohexyl: $\text{C}_6\text{H}_{11}-\text{MgBr}$.



Worked example · NCERT Example 7.2 — products of three preparations

1. Catalytic reduction of butanal $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$: an aldehyde gives a primary alcohol, **butan-1-ol**.
2. Hydration of propene with dilute H_2SO_4 : Markovnikov, **propan-2-ol**.
3. Propanone CH_3COCH_3 with CH_3MgBr then water: a ketone gives a tertiary alcohol, $(\text{CH}_3)_2\text{C}(\text{OH})\text{CH}_3$ = **2-methylpropan-2-ol**.

Worked example · NCERT intext 7.5 — products with H_3O^+ and with NaBH_4

1. Propene with H_3O^+ : acid hydration, **propan-2-ol**.
2. The cyclohexanone that has a $\text{CH}_2\text{COOCH}_3$ side chain, with NaBH_4 : the ketone becomes a secondary alcohol and the **ester stays untouched** (mild reagent).
3. 2-Methylbutanal $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CHO}$ with NaBH_4 : aldehyde to primary alcohol, **2-methylbutan-1-ol**.

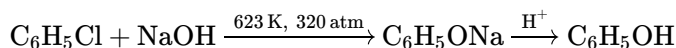
Examiner asks: "write the product of (a ketone or ester compound) with NaBH_4 " (2018); "a Grignard reagent with an aldehyde or ketone, then hydrolysis" (2024); "convert propanone to 2-methylpropan-2-ol" and "convert butan-2-one to butan-2-ol". Use the table above and name which reagent leaves the ester alone.

Preparing phenols — four routes — three lab, one industrial

Phenol is also called **carbolic acid**. It was first isolated in the early nineteenth century from **coal tar** (the sticky by-product of heating coal). Today it is made synthetically, and most of the world's phenol comes from cumene.

1. From a haloarene (chlorobenzene)

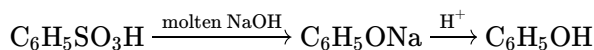
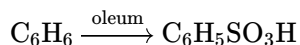
Chlorobenzene is **fused** with sodium hydroxide (heated molten with it) at **623 K and 320 atm** pressure. This gives **sodium phenoxide** $\text{C}_6\text{H}_5\text{ONa}$, a salt; acidifying it releases phenol.



2. From benzenesulphonic acid

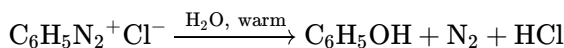
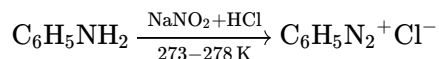
[exposure] **Oleum** is fuming sulphuric acid, $\text{H}_2\text{S}_2\text{O}_7$ (concentrated H_2SO_4 with dissolved SO_3).

Sulphonation puts an $-\text{SO}_3\text{H}$ group on the ring. Benzene is sulphonated with oleum to give **benzenesulphonic acid**; heated with **molten sodium hydroxide** this becomes sodium phenoxide, and acidification gives phenol.



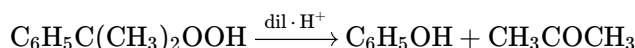
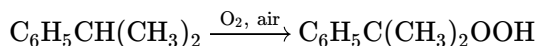
3. From a diazonium salt

[exposure] A **diazonium salt** ArN_2^+X^- is made by treating an **aromatic primary amine** (an $-\text{NH}_2$ directly on a benzene ring, such as aniline) with **nitrous acid**, which is made in the flask from $\text{NaNO}_2 + \text{HCl}$, at **273–278 K**. The $\text{C}-\text{N}_2^+$ bond is weak, so the N_2^+ is easy to displace. **Warming with water** (or treating with dilute acid) hydrolyses it to phenol and releases nitrogen gas.



4. From cumene (the industrial route)

[exposure] **Cumene** is isopropylbenzene, $\text{C}_6\text{H}_5-\text{CH}(\text{CH}_3)_2$: a benzene ring on the middle carbon of propane. It is **oxidised by the oxygen of air** to **cumene hydroperoxide**, in which the benzylic carbon carries an $-\text{O}-\text{O}-\text{H}$ group. Treating that with **dilute acid** splits it into **phenol and acetone**.



This is the preferred commercial method for two reasons: it starts from a cheap hydrocarbon, and the by-product **acetone** is itself an important compound obtained in large quantity. Two useful products from one

process.

Examiner asks: "preparation of phenol from cumene" (2025), "convert chlorobenzene to phenol" (2019) and "convert aniline to phenol" (year not stated). Write the reagent and condition over each arrow. The chlorobenzene route needs the high temperature and pressure.

Physical properties: boiling point and solubility — *one explanation — hydrogen bonding through the O–H — covers both*

Alcohols and phenols have two parts: an alkyl or aryl group, and the hydroxyl. The properties come chiefly from the –OH; the carbon group only modifies them.

[exposure] A **hydrogen bond** is an attraction between a hydrogen that is bonded to a very electronegative atom (here oxygen) and a lone pair on a neighbouring oxygen. The oxygen pulls the shared electrons, so the oxygen carries δ^- and the hydrogen carries δ^+ , and the δ^+ hydrogen of one molecule is attracted to the δ^- oxygen of the next. **Intermolecular** means *between* separate molecules; **intramolecular** means *inside* one molecule. It is much stronger than the ordinary **van der Waals forces**, the weak attractions that exist between any two molecules and grow with surface area.

Boiling point

- **Rises with the number of carbons:** a bigger molecule has more surface contact and larger van der Waals forces.
- **Falls with branching:** a branched molecule is more compact, so less surface area and weaker van der Waals forces. Butan-2-ol boils lower than butan-1-ol.
- **Alcohols and phenols boil much higher than hydrocarbons, ethers, haloalkanes and haloarenes of similar molar mass**, because the –OH forms **intermolecular hydrogen bonds** that those others lack. Boiling needs the molecules to be pulled apart, so these bonds must be broken first.

Compound	Type	Molar mass	b.p. (K)
Ethanol C ₂ H ₅ OH	alcohol	46	351
Methoxymethane CH ₃ OCH ₃	ether	46	248
Propane C ₃ H ₈	hydrocarbon	44	231

Same size, three very different boiling points; only the alcohol can hydrogen-bond *to itself*.

Solubility in water

Alcohols and phenols dissolve in water because they form hydrogen bonds with water. The –OH hydrogen is attracted to a water oxygen, and a water hydrogen to the –OH oxygen. The carbon group is **[exposure] hydrophobic** ("water-hating": non-polar, it does not attract water and gets in the way of hydrogen bonding), so **solubility falls as the alkyl or aryl group gets bigger**. **[exposure]** Two liquids are **miscible** if they mix in all proportions; methanol and ethanol are miscible with water in all proportions, while liquids that do not mix are **immiscible**.

Worked example · NCERT Example 7.3 — arrange in increasing boiling point

(a) pentan-1-ol, butan-1-ol, butan-2-ol, ethanol, propan-1-ol, methanol. (b) pentan-1-ol, n-butane, pentanal, ethoxyethane.

- (a) All are alcohols, so rank by carbon count, then branching: methanol (1 C), ethanol (2), propan-1-ol (3). Butan-2-ol and butan-1-ol both have 4 carbons; the branched butan-2-ol is lower. Then pentan-1-ol (5) is highest.
- (b) n-Butane is non-polar, only weak van der Waals forces: lowest. Ethoxyethane is an ether, weakly polar. Pentanal is an aldehyde with a strongly polar C=O and stronger dipole–dipole attraction. Pentan-1-ol hydrogen-bonds: highest.

Answer: (a) methanol < ethanol < propan-1-ol < butan-2-ol < butan-1-ol < pentan-1-ol. (b) n-butane < ethoxyethane < pentanal < pentan-1-ol.

Examiner asks: "why does propanol have a higher boiling point than butane?" (2019), the boiling-point arrangement of four compounds (ethoxyethane, butanal, butanol, butane; 2023). NCERT exercise 7.22 asks the same thing of ethanol against methoxymethane. The fixed answer: hydrogen bonding.

Two ways an alcohol reacts — *which bond breaks sorts the whole reaction list*

An alcohol is **versatile**: it can behave as a nucleophile and also, once protonated, as an electrophile.

- As a nucleophile:** the oxygen has lone pairs, so it is electron-rich. It attacks a positive carbon, and in the process the **O–H bond breaks** (the hydrogen leaves as H^+).
- As an electrophile** (only after it picks up H^+): the oxygen takes a proton from an acid and becomes a **[exposure] protonated alcohol** ROH_2^+ , also called an **alkyloxonium ion**. The positive oxygen drags electrons away from the carbon, so the carbon now carries $\delta+$ and a nucleophile can attack it; water leaves as the leaving group, so the **C–O bond breaks**. A plain alcohol does not act as an electrophile; only the protonated one does.

On that basis the reactions fall into two lists, and the rest of the chapter follows it:

O–H bond breaks	C–O bond breaks (alcohols only)
Reaction with metals (acidity)	
Reaction with NaOH (phenols only)	
Esterification and acetylation	Reaction with HX and the Lucas test
Reaction with phosphorus halides and $SOCl_2$	
Dehydration to alkenes	
(Oxidation breaks O–H and C–H, a separate case)	

Phenols show C–O cleavage only with zinc dust, because the carbon of a phenol is sp^2 and the C–O bond is strong (partial double-bond character).

Examiner asks: nothing separately, but it organises every answer: before writing a mechanism ask "which bond is breaking?"

Acidity of alcohols and phenols — *proton donors; the weaker the conjugate base's grip on a negative charge, the stronger the acid*

[exposure] A **Brønsted acid** is a species that **donates a proton** (H^+) to a base. The **conjugate base** is what is left after it gives up the proton: ROH gives RO^- . A **stronger acid** gives up its proton more easily,

which happens when the conjugate base is **more stable**. Two things make an acid strong: the O–H bond is more polar (so the hydrogen leaves easily), and the negative charge left on oxygen is spread out and stabilised.

Reactions that show the acidity

- **With active metals** (sodium, potassium, aluminium: metals that displace hydrogen): alcohols give **alkoxides**, phenols give **phenoxides**, and hydrogen gas escapes. $2\text{ROH} + 2\text{Na} \longrightarrow 2\text{RONa} + \text{H}_2$; $2\text{C}_2\text{H}_5\text{OH} + 2\text{Na} \longrightarrow 2\text{C}_2\text{H}_5\text{ONa} + \text{H}_2$ (sodium ethoxide); $2\text{C}_6\text{H}_5\text{OH} + 2\text{Na} \longrightarrow 2\text{C}_6\text{H}_5\text{ONa} + \text{H}_2$.
- **With aqueous NaOH — phenols only:** $\text{C}_6\text{H}_5\text{OH} + \text{NaOH} \longrightarrow \text{C}_6\text{H}_5\text{ONa} + \text{H}_2\text{O}$, an acid-plus-base neutralisation to a salt and water. Alcohols do **not** do this, which is the evidence that **phenols are more acidic than alcohols and water**.

Acidity of alcohols: weak, and the carbon group makes it weaker

The acidity of an alcohol comes from the polar O–H bond: oxygen is electronegative, so oxygen is δ^- and hydrogen δ^+ , and the electrons are held towards oxygen. An alkyl group is an **[exposure] electron-releasing** group (it pushes electron density towards the atom it is attached to; this is the +I effect). It pushes electrons *onto the oxygen*, which lowers the polarity of O–H, so the hydrogen leaves less easily. More alkyl groups, less acidity:

Acid strength, strongest first: methanol > primary > secondary > tertiary alcohols.

A primary alcohol has one pushing alkyl group, a secondary has two and a tertiary three. Alcohols are also **weaker acids than water**: water has no alkyl group to push electrons. Proof: an alkoxide takes a proton from water, $\text{RO}^- + \text{H}_2\text{O} \longrightarrow \text{ROH} + \text{OH}^-$, so water is the better proton donor, and an **alkoxide ion is a stronger base than hydroxide** (sodium ethoxide is a stronger base than sodium hydroxide). Alcohols are also **Brønsted bases** as well as acids: the lone pairs on oxygen let them *accept* a proton.

Acidity of phenol: why it beats alcohols and water

[exposure] Resonance is the delocalisation of electrons over several atoms, so that the true structure is a blend of several drawings (**resonance structures**) that differ only in where the electrons sit. A charge spread over several atoms is more stable than one stuck on a single atom. Phenol is acidic for two reasons.

1. **The –OH carbon is sp^2 , not sp^3 .** An sp^2 orbital has 33% s-character against 25% for sp^3 , and more s-character means the carbon holds its electrons more tightly (**more electronegative**). So in phenol the carbon pulls electron density off the oxygen, the O–H bond becomes more polar, and the proton leaves more easily. In an alcohol the alkyl carbon is less electronegative and also pushes electrons onto oxygen.
2. **Phenoxide ion is stabilised by resonance; alkoxide ion is not.** In RO^- the whole negative charge is stuck on one oxygen atom — concentrated charge, unstable. In $\text{C}_6\text{H}_5\text{O}^-$ the negative charge spreads from oxygen onto the ortho and para carbons of the ring (five resonance structures), so it is spread out and the ion is far more stable. A more stable conjugate base means a stronger acid.

Phenol itself also has resonance structures, but they all have **charge separation** (a plus on oxygen and a minus on a ring carbon), which is destabilising; so the phenoxide ion gains much more from resonance than phenol does. That difference of stability is what drives ionisation. Phenol is about a million times more acidic than ethanol (pK_a 10.0 against 15.9).

Substituted phenols

[exposure] pK_a is a number that measures acid strength: **the smaller the pK_a , the stronger the acid** ($\text{pK}_a = -\log K_a$). **[exposure]** An **electron-withdrawing group** (such as $-\text{NO}_2$) pulls electron

density towards itself; an electron-releasing group (such as $-\text{CH}_3$) pushes it away.

- **Electron-withdrawing groups ($-\text{NO}_2$) increase acidity**, most strongly at **ortho and para**, because the negative charge of the phenoxide ion can be delocalised right onto the nitro group.
- **Electron-releasing groups ($-\text{CH}_3$, alkyl) decrease acidity**: cresols are less acidic than phenol.

Compound	pK _a	Compound	pK _a
o-Nitrophenol	7.2	o-Cresol	10.2
m-Nitrophenol	8.3	m-Cresol	10.1
p-Nitrophenol	7.1	p-Cresol	10.2
Phenol	10.0	Ethanol	15.9

Read the table: p- and o-nitrophenol (7.1, 7.2) are stronger acids than m-nitrophenol (8.3), because the meta nitro cannot take the negative charge by resonance. All three nitrophenols are stronger than phenol (10.0), and the cresols (about 10.2) are weaker.

Worked example · NCERT intext 7.8 — why o- and p-nitrophenol are more acidic than phenol

1. Remove H^+ from o- or p-nitrophenol to get the nitrophenoxide ion. As in phenoxide, the negative charge on oxygen moves onto the ring carbons ortho and para to the oxygen.
2. The nitro group is itself at an ortho or para carbon. Charge arriving there can move on into the $\text{N}=\text{O}$ part of the nitro group, ending up on its oxygen atoms. That is two more resonance structures than plain phenoxide has.
3. The charge is delocalised over the ring **and** onto the nitro group; phenoxide has only the ring. A more stable conjugate base means a stronger acid.

Worked example · NCERT Example 7.4 — increasing acid strength

Propan-1-ol, 2,4,6-trinitrophenol, 3-nitrophenol, 3,5-dinitrophenol, phenol, 4-methylphenol.

1. The alcohol is weakest of all (no resonance in its conjugate base).
2. Among the phenols, the methyl (electron-releasing) is weaker than phenol itself.
3. Then nitrophenols, in order of the number of nitro groups: one, two, three. 2,4,6-Trinitrophenol (picric acid) has three nitro groups, two of them at ortho and para, so it is the strongest.

Answer: propan-1-ol < 4-methylphenol < phenol < 3-nitrophenol < 3,5-dinitrophenol < 2,4,6-trinitrophenol.

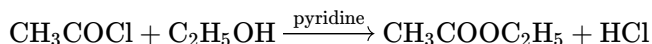
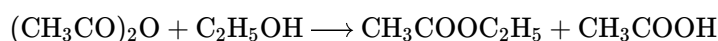
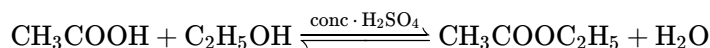
Trap: an alcohol is **not** a stronger acid because "alkyl groups are bigger" — the order of alcohol acidity is primary > secondary > tertiary because alkyl groups push electrons *onto* the oxygen, which makes the proton harder to lose. And for phenols the nitro group helps most at ortho and para; at meta it helps less.

Examiner asks: "why is phenol more acidic than cyclohexanol" (2023) and "why is an alcohol less acidic than phenol" (2019): resonance-stabilised phenoxide against an unstabilised alkoxide; "arrange p-cresol, p-nitrophenol and phenol" (2017); "ortho and para nitrophenols are more acidic than phenol, draw the resonance structures". Always give the conjugate-base argument.

Esterification and acetylation — *the O–H bond breaks; the oxygen attacks an acid derivative*

[exposure] An **ester** is a compound RCOOR' made from a carboxylic acid and an alcohol (or phenol). **Esterification** is the making of an ester. The group $\text{RCO}-$ (the acid with its $-\text{OH}$ removed) is called the **[exposure] acyl group**; it combines with the $\text{R}'\text{O}-$ part of the alcohol. Alcohols and phenols give esters with three kinds of acid derivative:

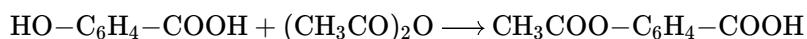
Reacts with	Conditions	By-product
Carboxylic acid RCOOH	a little conc. H_2SO_4 (catalyst, and it removes water)	H_2O
Acid anhydride $(\text{RCO})_2\text{O}$	a little conc. H_2SO_4	RCOOH
Acid chloride RCOCl	in the presence of a base, pyridine	HCl (taken up by the pyridine)



All three give **ethyl ethanoate**. **[exposure]** The reaction with the acid is **reversible** (it runs in both directions until an equilibrium is reached). To make more ester, **remove the water as soon as it forms**: by Le Chatelier's principle, taking away a product pushes the equilibrium to the right. With the acid chloride, the base neutralises the HCl that forms, which also shifts the equilibrium to the right.

Acetylation and aspirin

[exposure] **Acetylation** is putting an **acetyl** group $\text{CH}_3\text{CO}-$ onto the oxygen of an alcohol or phenol. Salicylic acid (2-hydroxybenzoic acid) has a phenolic $-\text{OH}$ and a $-\text{COOH}$. Acetic anhydride acetylates the $-\text{OH}$ and leaves the $-\text{COOH}$ alone: the product is **acetylsalicylic acid, aspirin**, with ethanoic acid as by-product.



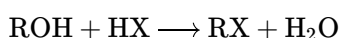
with CH_3COOH as the by-product.

Aspirin is used as a medicine: it is an **analgesic** (pain reliever), **anti-inflammatory** and **antipyretic** (fever reducer).

Examiner asks: acetylation of salicylic acid to give aspirin (2020, "name the product and its use"); why pyridine is used with an acid chloride. Say: it neutralises HCl and shifts the equilibrium forward.

Reactions of alcohols that break the C–O bond: halides and the Lucas test — *protonate the oxygen, let water leave, let a halide in*

1. With hydrogen halides (and the Lucas test)



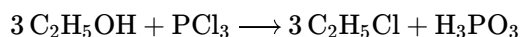
The alcohol is protonated (the oxygen takes H^+), water leaves, and the halide ion attacks the carbon. The three classes of alcohol react at **very different speeds**, and that is how they are told apart. **[exposure]** The **Lucas reagent** is a mixture of **concentrated** HCl and **anhydrous** ZnCl_2 . All the alcohols start by dissolving in it. As the alcohol turns into an alkyl chloride, the chloride is **immiscible** (does not mix) with the solution, so it separates as tiny droplets and the solution goes cloudy. That cloudiness is **[exposure]** **turbidity**.

Alcohol class	What you see with Lucas reagent at room temperature
Tertiary	turbidity immediately : fastest
Secondary	turbidity after about five minutes
Primary	no turbidity at room temperature (it reacts only on heating)

Reactivity: **tertiary** > **secondary** > **primary**, because a tertiary alcohol forms its more stable carbocation fastest. A primary alcohol, such as propan-1-ol, gives no cloudiness in the cold; 2-methylpropan-2-ol gives it at once. That pair is a favourite question.

2. With phosphorus trihalides and thionyl chloride

Alcohols react with PCl_3 , PBr_3 and PI_3 to give alkyl halides; the phosphorus ends up as phosphorous acid H_3PO_3 . With **thionyl chloride** SOCl_2 the OH is replaced by Cl, and the by-products are both gases, SO_2 and HCl , which escape and make the chloride easy to isolate.



Worked example · NCERT intext 7.6 — butan-1-ol and 2-methylbutan-2-ol with (a) HCl/ZnCl_2 , (b) HBr , (c) SOCl_2

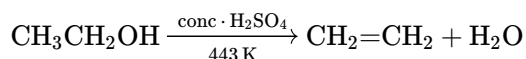
In every case $-\text{OH}$ is replaced by the halogen; the carbon skeleton does not change.

1. Butan-1-ol, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$: (a) 1-chlorobutane (slowly, primary); (b) **1-bromobutane**; (c) **1-chlorobutane** + SO_2 + HCl .
2. 2-Methylbutan-2-ol, $\text{CH}_3\text{CH}_2\text{C}(\text{CH}_3)_2\text{OH}$: (a) **2-chloro-2-methylbutane** (immediately, tertiary); (b) **2-bromo-2-methylbutane**; (c) 2-chloro-2-methylbutane.

Examiner asks: the product of butan-2-ol with SOCl_2 (2-chlorobutane, 2017); "distinguish 1-propanol from 2-methyl-2-propanol" (2019); Lucas test; HCl with a compound that has both a phenolic and a benzylic $-\text{OH}$ (2020, see PYQ).

Dehydration of alcohols to alkenes — *remove water from one molecule; the temperature and acid strength depend on the class*

[exposure] **Dehydration** means removing a molecule of water. Heated with a **[exposure]** **protic acid** (an acid that gives up H^+ : concentrated H_2SO_4 , or H_3PO_4), or over catalysts such as **anhydrous** ZnCl_2 or **alumina** Al_2O_3 , an alcohol loses OH from one carbon and H from the neighbouring carbon and gives an **alkene**.



The class of the alcohol sets how hard you must push. **Primary** needs strong acid and high temperature; **secondary and tertiary** dehydrate under **milder conditions**:

Alcohol	Conditions	Product
Ethanol (1°)	conc. H ₂ SO ₄ , 443 K	ethene
Propan-2-ol (2°)	85% H ₃ PO ₄ , 440 K	propene
tert-Butyl alcohol (3°)	20% H ₃ PO ₄ , 358 K	2-methylpropene

Relative ease of dehydration: tertiary > secondary > primary. The reason is the same as in the Lucas test: the step that decides the speed forms a carbocation, and tertiary carbocations are the most stable and so form most readily.

Mechanism (ethanol to ethene), three steps

- Formation of protonated alcohol** (fast). The oxygen has lone pairs and takes H⁺ from the acid: $\text{CH}_3\text{CH}_2\text{OH} + \text{H}^+ \longrightarrow \text{CH}_3\text{CH}_2\text{OH}_2^+$ (an **ethyloxonium ion**; **[exposure]** an **oxonium ion** is an oxygen carrying a positive charge and three bonds).
- Formation of carbocation (slowest step)**. The positive oxygen pulls the bonding electrons towards itself, the C–O bond breaks and water leaves: $\text{CH}_3\text{CH}_2\text{OH}_2^+ \longrightarrow \text{CH}_3\text{CH}_2^+ + \text{H}_2\text{O}$. Because it is the slowest step of the reaction, it is the **rate-determining step**: the overall speed is the speed of this step.
- Formation of ethene by elimination of a proton**. The carbocation loses H⁺ from the neighbouring carbon; the C–H electrons form the C=C: $\text{CH}_3\text{CH}_2^+ \longrightarrow \text{CH}_2=\text{CH}_2 + \text{H}^+$.

The acid used in step 1 is released in step 3, so it behaves as a catalyst. The reaction is an equilibrium, so to push it to the right **ethene is removed as it forms** (Le Chatelier's principle again).

Worked example · NCERT intext 7.7 — major product of acid-catalysed dehydration

- 1-Methylcyclohexanol**: a tertiary alcohol. OH leaves from the ring carbon that carries the methyl, and an H leaves from a neighbouring ring carbon: **1-methylcyclohexene**, the alkene with three alkyl groups on the double bond (the most stable).
- Butan-1-ol**: gives a **mixture of but-1-ene and but-2-ene**, with **but-2-ene as the major product**. It has two alkyl groups on the C=C and is more stable than but-1-ene; NCERT's explanation is that the primary carbocation **rearranges** (a hydride shift) to the more stable secondary carbocation before losing H⁺.

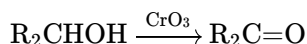
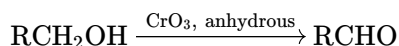
Trap: "which dehydrates faster, tert-butyl alcohol or n-butanol?" The answer is **tert-butyl alcohol**, and the argument must be about the **carbocation** (tertiary is more stable, so the slow step is faster) — not just "tertiary has more carbons". Write the protonation and the carbocation for both and compare.

Examiner asks: the mechanism of dehydration of ethanol to ethene at 443 K (2019); "tert-butyl alcohol heated with 20% H₃PO₄ at 358 K" (2020); "which undergoes dehydration faster" (2020). Always give the three steps with their names.

Oxidation of alcohols — *primary, secondary and tertiary alcohols give three different outcomes — choose the reagent*

[exposure] In organic chemistry **oxidation** of an alcohol means forming a C=O bond at the carbon that carried the –OH, which needs **that carbon to have a hydrogen to lose**, together with the hydrogen of the O–H. It is also called **dehydrogenation** (loss of H₂). The oxidising agent decides how far a primary alcohol goes.

Alcohol	Oxidising agent	Product
Primary	Strong: acidified KMnO ₄ or K ₂ Cr ₂ O ₇	aldehyde first, which is oxidised further to a carboxylic acid with the same number of carbons
Primary	Mild: CrO ₃ in anhydrous medium, or PCC	stops at the aldehyde
Secondary	CrO ₃ (chromic anhydride), K ₂ Cr ₂ O ₇ , PCC	ketone with the same number of carbons
Tertiary	normal oxidants	no reaction



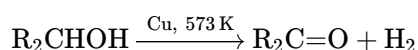
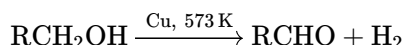
[exposure] PCC is **pyridinium chlorochromate**, a complex of chromium trioxide with pyridine and HCl. It is the **better reagent** for primary alcohol to aldehyde: good yield of aldehyde and it does not touch a C=C double bond in the molecule. Why do we need a mild reagent? Because an aldehyde is easier to oxidise than the alcohol; a strong oxidant would carry it on to the acid.

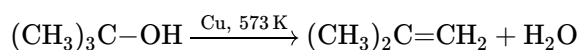
Tertiary alcohols resist oxidation

The carbon that carries –OH has no hydrogen, so no C=O can form without breaking a **C–C bond**, which is hard. Under severe conditions (strong oxidant such as KMnO₄ and high temperature) C–C bonds do break, and you get a **mixture of carboxylic acids with fewer carbons**.

Catalytic dehydrogenation

When the **vapour** of a primary or secondary alcohol is passed over **heated copper at 573 K**, hydrogen gas is removed: a primary alcohol gives an **aldehyde**, a secondary alcohol a **ketone**. A tertiary alcohol has no hydrogen on the –OH carbon, so under the same conditions it **dehydrates to an alkene** instead (it loses OH and a β hydrogen).





Why methanol is poisonous

In the body methanol is oxidised first to methanal and then to methanoic acid, which can cause **blindness and death**. Patients are treated with intravenous dilute ethanol: ethanol uses up the same enzyme, giving the kidneys time to excrete the methanol.

Examiner asks: "butan-2-ol with CrO_3 " (2017: butanone); "ethanol to ethanal" (2019: PCC, not KMnO_4); "tert-butyl alcohol over copper at 573 K" (2020: 2-methylpropene); "why are tertiary alcohols resistant to oxidation" (2025). Match the reagent to the class before you write anything.

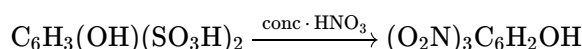
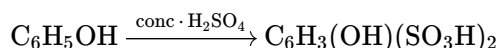
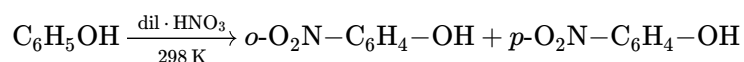
Reactions of phenol on the ring — *–OH makes the ring electron-rich, so it reacts readily and at the ortho and para positions*

[exposure] Electrophilic aromatic substitution is a reaction in which an electron-poor species (an electrophile) replaces a hydrogen on a benzene ring. A group already on the ring either **activates** it (makes it react faster than benzene) or **deactivates** it, and **directs** the new group to certain positions.

The $-\text{OH}$ in phenol **activates** the ring and directs the new group to the **ortho and para positions**. The oxygen has a lone pair that it can push into the ring by **resonance** (the +R effect). The lone pair pushes electrons onto the carbons ortho and para to the $-\text{OH}$ (two ortho carbons and one para), so these positions are electron-rich and attract an electrophile.

1. Nitration

- **Dilute HNO_3 at low temperature (298 K):** a mixture of **o-nitrophenol** and **p-nitrophenol**.
- **Concentrated HNO_3 :** three nitro groups (two ortho, one para): **2,4,6-trinitrophenol, picric acid**. The yield is poor. The better way: phenol with conc. H_2SO_4 gives phenol-2,4-disulphonic acid, which is then nitrated with conc. HNO_3 ; the sulphonic acid groups are replaced and picric acid forms in good yield. Picric acid is a strong acid because three electron-withdrawing nitro groups spread the negative charge.

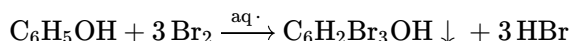
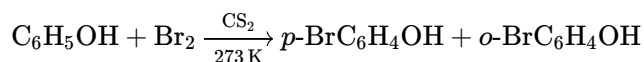


The last product is 2,4,6-trinitrophenol, picric acid.

[exposure] Steam distillation separates liquids by boiling them with steam: a compound that is **steam volatile** (easily carried over as vapour with steam) comes over first. Ordinary ortho and para isomers are hard to separate, but here you can: **o-nitrophenol is steam volatile** because it has an **intramolecular hydrogen bond** (between the $-\text{OH}$ and the neighbouring $-\text{NO}_2$ of the *same* molecule), so its molecules do not stick to each other and vaporise easily. **p-Nitrophenol is less volatile:** its hydrogen bonds are **intermolecular** (the $-\text{OH}$ of one molecule bonds to the $-\text{NO}_2$ of another), so molecules are **associated**, and heating must first break these bonds before they vaporise.

2. Halogenation

- Br₂ in a solvent of low polarity (CS₂ or CHCl₃) at low temperature (273 K): **monobromophenols**, p-bromophenol the major product and o-bromophenol the minor one.
- **Bromine water: 2,4,6-tribromophenol**, a **white precipitate**. Water is very polar, so it polarises Br₂ further and all three positions react.



The precipitate is 2,4,6-tribromophenol.

Benzene needs a Lewis acid (FeBr₃) to polarise Br₂ into Br⁺; **phenol does not need it**, because the ring is so electron-rich that an approaching Br₂ is polarised by the ring itself: **the highly activating effect of –OH**.

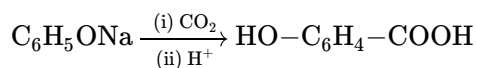
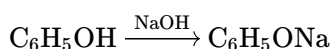
Worked example · NCERT Example 7.5 — nitration of 3-methylphenol

Put –OH at C-1 and –CH₃ at C-3. The –OH directs to C-2, C-4 and C-6; the methyl (also ortho/para-directing) directs to C-2, C-4 and C-6 as well, so they agree. C-2 is between the two groups and too crowded for the incoming group, so the nitro group goes to C-4 or C-6.

1. Mononitration: **3-methyl-4-nitrophenol** and **5-methyl-2-nitrophenol**.
2. Dinitration: both C-4 and C-6: **5-methyl-2,4-dinitrophenol**.
3. Mononitration of phenyl methanoate (the ester C₆H₅OCHO): the ester oxygen also donates a lone pair, so the group is ortho/para-directing, and the major product is the para nitro compound.

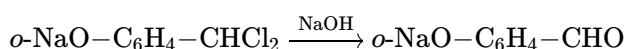
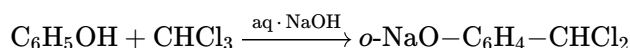
3. Kolbe's reaction

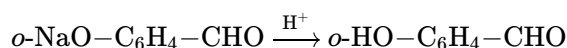
Phenol with NaOH gives **sodium phenoxide**, which is **even more reactive** than phenol towards electrophiles, because the ion carries a full negative charge pushing into the ring. That is enough for it to react with **carbon dioxide**, which is only a **weak electrophile**. CO₂ goes to the ortho position, and acidification gives **2-hydroxybenzoic acid, salicylic acid**.



4. Reimer–Tiemann reaction

Phenol is treated with **chloroform** CHCl₃ in **aqueous** NaOH. A –CHCl₂ group goes onto the ortho position (the phenol is present as the phenoxide), giving an intermediate **substituted benzal chloride**; the alkali hydrolyses this to a –CHO group, and acidification gives **2-hydroxybenzaldehyde, salicylaldehyde**. So the –CHO group is introduced at the ortho position.





5. Zinc dust, oxidation, and the iron(III) test

- **With zinc dust**, heated: phenol is reduced to **benzene**, and zinc is oxidised to ZnO.
 $\text{C}_6\text{H}_5\text{OH} + \text{Zn} \longrightarrow \text{C}_6\text{H}_6 + \text{ZnO}$. This is the only C–O cleavage a phenol shows.
- **Oxidation** with chromic acid ($\text{Na}_2\text{Cr}_2\text{O}_7$ with H_2SO_4) gives **benzoquinone**, a **[exposure] conjugated diketone** (a ring with two C=O groups and two C=C in alternation; $\text{O}=\text{C}_6\text{H}_4=\text{O}$). In air, phenol slowly oxidises to a dark mixture containing quinones.
- **Neutral FeCl_3 test** (asked in board papers): phenol gives a **violet colour** (an iron–phenoxide complex, $[\text{Fe}(\text{OC}_6\text{H}_5)_6]^{3-}$); alcohols give nothing. It is the standard way to tell a phenol from an alcohol. This test is not printed on the NCERT pages checked for these notes; it comes from the PYQ video.

Examiner asks: "write the Reimer–Tiemann and Kolbe reactions" (2016), "phenol to picric acid" (year not stated), "phenol to benzene / toluene" (2019), "why is o-nitrophenol steam volatile" (2019), "phenol with bromine water" (2024), "phenol with bromine in CS_2 " (2024). Name the product and say what it looks like (white precipitate; yellow solid for picric acid).

Methanol and ethanol, the two commercial alcohols — *where they come from, the numbers, and why one is a poison*

Methanol, CH_3OH

- Called **wood spirit** because it was once made by the **[exposure] destructive distillation of wood** (heating wood strongly in the absence of air so that it breaks down).
- Today it is made by **catalytic hydrogenation of carbon monoxide** at high temperature and pressure over a $\text{ZnO}-\text{Cr}_2\text{O}_3$ catalyst: $\text{CO} + 2\text{H}_2 \xrightarrow[573-673\text{ K, } 200-300\text{ atm}]{\text{ZnO}-\text{Cr}_2\text{O}_3} \text{CH}_3\text{OH}$. (The numeric conditions are quoted in the lecture; NCERT says only "high pressure and temperature".)
- Colourless liquid, boiling point **337 K**, **highly poisonous**: small quantities cause blindness, large ones death.
- Used as a solvent in paints and varnishes, and chiefly for making methanal (formaldehyde).

Ethanol, $\text{C}_2\text{H}_5\text{OH}$

[exposure] Fermentation is the breaking down of sugar by the action of yeast, in the **absence of air** (**anaerobic** conditions), giving alcohol and carbon dioxide. An **[exposure] enzyme** is a protein that speeds up a reaction in a living thing; each enzyme works on one kind of reaction.

1. The sugar in molasses, sugarcane or fruits such as grapes is **sucrose**. The enzyme **invertase** hydrolyses it into **glucose and fructose**, both of formula $\text{C}_6\text{H}_{12}\text{O}_6$ (same formula, different structures).
2. The enzyme **zymase**, present in **yeast**, ferments glucose or fructose:

$$\text{C}_6\text{H}_{12}\text{O}_6 \xrightarrow{\text{zymase}} 2\text{C}_2\text{H}_5\text{OH} + 2\text{CO}_2$$

In wine making, grapes supply both sugar and yeast: as grapes ripen the sugar content rises and yeast grows on the outer skin; when grapes are crushed, the sugar and the yeast enzyme meet and fermentation starts.

- **The action of zymase stops once the alcohol exceeds about 14%.** To get stronger alcohol you must **distil**.

- If **air gets into the fermentation mixture**, its oxygen oxidises ethanol to **ethanoic acid**, which destroys the taste of the drink.
- Ethanol is a colourless liquid, boiling point **351 K**. Uses: a solvent in the paint industry, and to make many carbon compounds. Today much ethanol is made by **hydration of ethene**.

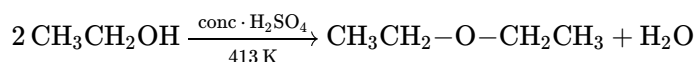
[exposure] Denatured alcohol: commercial ethanol is made unfit to drink by adding **copper sulphate** (to give it a colour) and **pyridine** (a foul-smelling liquid), so that nobody drinks it. This is called **denaturation** of alcohol.

Examiner asks: recall questions: the catalyst for methanol ($\text{ZnO}-\text{Cr}_2\text{O}_3$), the enzymes invertase and zymase, why fermentation stops at 14%, and what denatured alcohol contains. Not a heavy section, but easy marks.

Preparing ethers — *two routes — dehydration of a primary alcohol, and the Williamson synthesis*

1. Dehydration of alcohols (bimolecular)

The same acid dehydration can give an alkene or an ether, depending on the conditions. With ethanol and conc. H_2SO_4 : at **443 K** the product is **ethene**; at the lower temperature of **413 K**, with excess alcohol, the product is **ethoxyethane**. This is called **bimolecular** dehydration because **two alcohol molecules** combine, losing one water molecule.



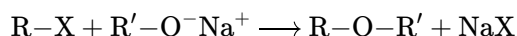
Mechanism (a nucleophilic bimolecular, **SN₂**, reaction: one step in which the nucleophile arrives while the leaving group departs):

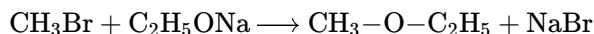
1. **Protonation:** $\text{CH}_3\text{CH}_2\text{OH} + \text{H}^+ \longrightarrow \text{CH}_3\text{CH}_2-\text{OH}_2^+$.
2. **Attack of a second alcohol molecule** on the protonated alcohol: its oxygen lone pair attacks the carbon (which is $\delta+$ because the oxygen is positive), and water leaves in the same step. This gives a protonated ether: $\text{CH}_3\text{CH}_2-\text{OH}_2^+ + \text{CH}_3\text{CH}_2\text{OH} \longrightarrow \text{CH}_3\text{CH}_2-\text{O}^+(\text{H})-\text{CH}_2\text{CH}_3 + \text{H}_2\text{O}$.
3. **Loss of a proton** from that oxygen gives the ether and returns H^+ : the acid is a catalyst.

Limits. The method works only for ethers with **primary alkyl groups**: the alkyl group must be unhindered (no branching) and the temperature kept low. With secondary and tertiary alcohols the carbon is crowded, one alcohol molecule cannot attack another, and **elimination wins over substitution**, so an alkene forms (tert-butyl alcohol with H_2SO_4 gives 2-methylpropene). The method also gives only **symmetrical** ethers, because it makes the same group on both sides. (NCERT asks why it is not appropriate for ethyl methyl ether: a mixture of alcohols gives a mixture of three ethers, CH_3OCH_3 , $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$ and $\text{CH}_3\text{OC}_2\text{H}_5$, not one product.)

2. Williamson synthesis

The important laboratory method, and the only one for **unsymmetrical** ethers (it also gives symmetrical ones). An **alkyl halide** reacts with a **sodium alkoxide**; the alkoxide ion is the nucleophile and displaces the halide by **SN₂**.



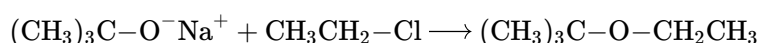


The **alkoxide may be primary, secondary or tertiary**, so an ether with a bulky group can be built, but **the alkyl halide must be primary**. An alkoxide is not only a nucleophile but also a **strong base**. With a secondary or tertiary halide it removes a hydrogen instead (**elimination**) and an alkene forms, with no ether at all.

Worked example · NCERT Example 7.6 — t-butyl ethyl ether

$(\text{CH}_3)_3\text{C—Cl} + \text{C}_2\text{H}_5\text{ONa}$ is "not an appropriate reaction" for making $(\text{CH}_3)_3\text{C—O—C}_2\text{H}_5$. What forms, and what is the right route?

1. The halide is **tertiary**: three methyl groups crowd the carbon and block the nucleophile (steric hindrance). Ethoxide therefore acts as a **base**, removes a hydrogen from a methyl group and the chloride leaves: **2-methylpropene** $(\text{CH}_3)_2\text{C=CH}_2$ is the major product.
2. Swap the roles: bulky group on the alkoxide, primary halide.



plus NaCl.

Worked example · NCERT intext 7.10 and 7.11

1. **Make 2-ethoxy-3-methylpentane from ethanol and 3-methylpentan-2-ol**. The ether has an ethyl on one side and a secondary group on the other. Primary halide, secondary alkoxide: make the alkyl halide from ethanol, $\text{C}_2\text{H}_5\text{OH} \xrightarrow{\text{HBr}} \text{C}_2\text{H}_5\text{Br}$, and the alkoxide from the secondary alcohol, $\text{R—OH} \xrightarrow{\text{Na}} \text{R—ONa}$; then combine: $\text{C}_2\text{H}_5\text{Br} + \text{R—ONa} \longrightarrow \text{R—O—C}_2\text{H}_5 + \text{NaBr}$.
2. **Which pair makes 1-methoxy-4-nitrobenzene: (i) 4-bromonitrobenzene + CH_3ONa , or (ii) sodium 4-nitrophenoxide + CH_3Br ?** Pair (ii). It uses a primary alkyl halide for $\text{S}_\text{N}2$. Pair (i) fails because an **aryl halide does not undergo nucleophilic substitution easily**: the C—Br bond has a partial double-bond character from resonance, and the carbon is sp^2 .

Phenols give ethers by the same method, with phenol as the phenoxide: phenol with NaOH gives sodium phenoxide, then CH_3Br gives **anisole**, $\text{C}_6\text{H}_5\text{OCH}_3$.

Examiner asks: the mechanism of ether formation from ethanol at 413 K (2023); the Williamson route to 2-methyl-2-methoxypropane (the alkyl halide must be CH_3Br , the alkoxide sodium tert-butoxide); "tert-butyl bromide with sodium methoxide gives an alkene, not an ether — give reason" (2019); phenol to anisole.

Physical properties of ethers — *polar bonds, but a boiling point like an alkane's*

The C—O bonds in an ether are polar (oxygen is more electronegative), so the molecule has a net **[exposure] dipole moment** (a measure of how unevenly charge is distributed). But the polarity is weak: the two alkyl groups sit either side of the oxygen and partly cancel it, so the ether is only weakly polar.

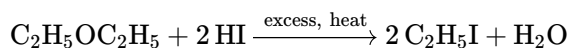
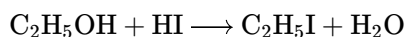
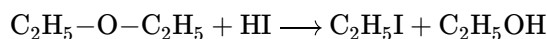
- **Boiling point** is comparable to that of an **alkane** of similar molar mass, and **much lower than an alcohol's**, because ethers cannot hydrogen-bond to each other (no O—H).
- **Solubility in water** is like that of an alcohol of the same molar mass: the ether oxygen can **accept a hydrogen bond from water**, even though the ether cannot hydrogen-bond to itself. An alkane cannot, and stays out of water.

Compound	Class	b.p. (K)	Solubility in 100 mL water
n-Pentane $\text{CH}_3(\text{CH}_2)_3\text{CH}_3$	alkane	309.1	essentially immiscible
Ethoxyethane $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$	ether	307.6	7.5 g
Butan-1-ol $\text{CH}_3(\text{CH}_2)_3\text{OH}$	alcohol	390	9 g

Examiner asks: NCERT exercise 7.22 (why is ethanol's boiling point higher than methoxymethane's) and the solubility comparison above ("can you explain this observation?" — the ether oxygen hydrogen-bonds with water, so ethoxyethane and butan-1-ol are about equally soluble while pentane is not).

Reactions of ethers: cleavage with HI — *ethers are the least reactive class — only a strong acid at high temperature breaks them*

Ethers are the **least reactive** of the functional groups here: no acidic hydrogen, weak polarity. The C–O bond breaks only under **drastic conditions: excess concentrated HI or HBr**, at high temperature. A dialkyl ether gives **two alkyl halide molecules** with excess acid. Order of reactivity of the hydrogen halides: **HI > HBr > HCl**, because HI is the strongest acid.

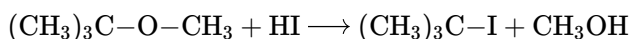


Mechanism for ethers with primary or secondary groups (SN2)

- Protonation** of the ether oxygen by HI: $\text{R--O--R}' + \text{HI} \longrightarrow \text{R--O}^+(\text{H})\text{--R}' + \text{I}^-$ (an oxonium ion). A protonated ether has a much better leaving group than the plain ether.
- Iodide attacks** and the alcohol leaves, in one step (SN2). Iodide is a good nucleophile and attacks the **less hindered, smaller** alkyl group (a big alkyl group blocks the approach and repels the iodide). So **the iodide ends up on the smaller group**: $\text{CH}_3\text{OC}_2\text{H}_5 + \text{HI} \longrightarrow \text{CH}_3\text{I} + \text{C}_2\text{H}_5\text{OH}$.
- With excess HI and heat, the alcohol formed reacts further to the alkyl iodide.

When one group is tertiary: SN1

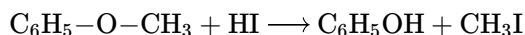
If one group is **tertiary**, the C–O bond to it breaks first, in a slow step, giving a **stable tertiary carbocation** (the other half leaves as alcohol); iodide then attacks the carbocation quickly. The mechanism is **SN1** (two steps), and the **tertiary alkyl iodide** forms.



Alkyl aryl ethers: phenol and an alkyl halide

In **anisole** $\text{C}_6\text{H}_5\text{--O--CH}_3$ the two C–O bonds differ. The O– C_6H_5 bond is **stronger**: the carbon is sp^2 , and the oxygen lone pair is shared with the ring, giving the bond **partial double-bond character**. So after protonation iodide attacks the **methyl** carbon (the O–CH₃ bond is weaker) and the products are **phenol**

and methyl iodide. The phenol does not react on to give an aryl halide, because the sp^2 carbon cannot undergo the nucleophilic substitution that would be needed.



Worked example · NCERT Example 7.7 — heating each ether with HI

1. $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{--O--CH}_2\text{CH}_3$ (both groups primary): SN_2 ; iodide takes the **smaller** group, ethyl: $\text{CH}_3\text{CH}_2\text{I} + \text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{OH}$.
2. $\text{CH}_3\text{CH}_2\text{CH}_2\text{--O--C}(\text{CH}_3)_2\text{CH}_2\text{CH}_3$ (one tertiary group): SN_1 ; tertiary iodide: $\text{CH}_3\text{CH}_2\text{C}(\text{CH}_3)_2\text{I} + \text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ (2-iodo-2-methylbutane and propan-1-ol).
3. $\text{C}_6\text{H}_5\text{--CH}_2\text{--O--C}_6\text{H}_5$ (benzyl phenyl ether): the aryl–O bond is the strong one; iodide takes the benzyl carbon: $\text{C}_6\text{H}_5\text{CH}_2\text{I} + \text{C}_6\text{H}_5\text{OH}$.

Worked example · NCERT intext 7.12 — products with HBr and HI

1. $\text{CH}_3\text{CH}_2\text{CH}_2\text{--O--CH}_3 + \text{HBr}$: both primary, SN_2 , bromide on the smaller (methyl) group: $\text{CH}_3\text{Br} + \text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$.
2. $(\text{CH}_3)_3\text{C--O--C}_2\text{H}_5 + \text{HI}$: one tertiary group, SN_1 : $(\text{CH}_3)_3\text{C--I} + \text{C}_2\text{H}_5\text{OH}$.

Trap: do not give the same answer for every ether. Check each side: **primary/secondary on both** → **iodide on the smaller group (SN_2)**; **a tertiary group present** → **the tertiary iodide (SN_1)**; **aryl on one side** → **phenol + alkyl iodide**, never an aryl iodide.

Examiner asks: "how can phenol be prepared from anisole? give reason" (2023); "ether with a tertiary butyl group heated with HI" (2025); "benzyl phenyl ether with HI" (2020). The reason for the aryl case is always the partial double bond.

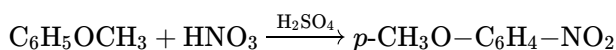
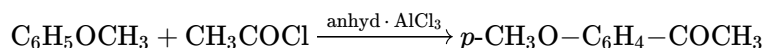
Reactions of aryl alkyl ethers on the ring — *–OR behaves like –OH: an activator that directs ortho and para*

An alkoxy group –OR on a benzene ring **activates** it and **directs ortho and para**, exactly as the –OH of phenol does, through the same resonance donation from the oxygen lone pair. The para product is the major one (anisole's para isomer is the favoured product: less crowded). Three reactions, all with anisole:

Reaction	Reagent	Major product	Minor product
Bromination	Br_2 in ethanoic acid, no FeBr_3 needed	p-bromoanisole (about 90%)	o-bromoanisole
Friedel–Crafts alkylation	CH_3Cl , anhydrous AlCl_3	4-methoxytoluene	2-methoxytoluene
Friedel–Crafts acylation	CH_3COCl , anhydrous AlCl_3	4-methoxyacetophenone	2-methoxyacetophenone
Nitration	conc. H_2SO_4 + conc. HNO_3	4-nitroanisole	2-nitroanisole

[exposure] A **Friedel–Crafts reaction** puts an **alkyl** group (alkylation) or an **acyl** group RCO-- (acylation) on a benzene ring, using an alkyl or acyl halide and a Lewis acid catalyst. A **[exposure] Lewis acid** is an electron-pair acceptor, such as AlCl_3 , which is electron-deficient. It pulls the chloride off the reagent: $\text{CH}_3\text{Cl} + \text{AlCl}_3 \longrightarrow \text{CH}_3^+ + \text{AlCl}_4^-$ and $\text{CH}_3\text{COCl} + \text{AlCl}_3 \longrightarrow \text{CH}_3\text{CO}^+ + \text{AlCl}_4^-$. The positive

CH_3^+ or CH_3CO^+ is the electrophile that the ring attacks. Acetophenone is a benzene ring with a COCH_3 group, which is a ketone.



Each reaction also gives a little of the ortho isomer.

Bromination needs no FeBr_3 for the same reason as in phenol: the methoxy group makes the ring so electron-rich that it polarises Br_2 itself.

Examiner asks: "Friedel–Crafts acylation of anisole" (2016, 3 marks), "nitration of anisole" (2025), "anisole with CH_3Cl and anhydrous AlCl_3 " (2024). Name both isomers and say which is major.

Patterns, collected — the orderings, tests, reagent choices and conversions you will actually be asked to produce

1. Rank by acidic strength

Worked example

Arrange in increasing acidic strength: p-cresol, p-nitrophenol, phenol (2017).

1. Look at the substituent: $-\text{CH}_3$ releases electrons, so p-cresol is weaker than phenol. $-\text{NO}_2$ withdraws electrons (and delocalises the charge from the para position), so p-nitrophenol is stronger.

Answer: p-cresol < phenol < p-nitrophenol.

- Increasing acidic strength: ethanol, phenol, water — answer only: ethanol < water < phenol
- Increasing acid strength of methanol, propan-2-ol, ethanol, 2-methylpropan-2-ol — answer only: 2-methylpropan-2-ol < propan-2-ol < ethanol < methanol
- Which is more acidic, o-nitrophenol or m-nitrophenol? — answer only: o-nitrophenol (pK_a 7.2 against 8.3); the ortho nitro group takes the charge by resonance

2. Rank by boiling point

Worked example

Increasing boiling point: ethoxyethane, butanal, butan-1-ol, n-butane (2023).

1. n-Butane: non-polar, only weak van der Waals forces: lowest.
2. Ethoxyethane: weakly polar; then butanal: strongly polar $\text{C}=\text{O}$, dipole–dipole attraction.
3. Butan-1-ol: hydrogen bonds, the strongest: highest.

Answer: n-butane < ethoxyethane < butanal < butan-1-ol.

- Higher boiling point: propan-1-ol or butane — answer only: propan-1-ol (hydrogen bonding)
- Higher boiling point: butan-1-ol or butan-2-ol — answer only: butan-1-ol (less branched, larger surface area)

3. Distinguish by a test

Worked example

Distinguish propan-1-ol from 2-methylpropan-2-ol (2019).

Add Lucas reagent (conc. HCl + anhydrous ZnCl_2) to each at room temperature. The tertiary alcohol, 2-methylpropan-2-ol, turns cloudy immediately (alkyl chloride is immiscible); the primary alcohol gives no turbidity at room temperature.

- Distinguish ethanol from dimethyl ether (2019) — answer only: iodoform test, $\text{I}_2 + \text{NaOH}$; ethanol (a $\text{CH}_3\text{CH}(\text{OH})-$ group) gives a yellow precipitate of CHI_3 , the ether gives none
- Distinguish phenol from cyclohexanol (2023) — answer only: neutral FeCl_3 ; phenol gives a violet colour, cyclohexanol none
- Tell primary, secondary and tertiary alcohols apart — answer only: Lucas test; turbidity immediate (3°), in about five minutes (2°), none at room temperature (1°)

4. Choose the oxidising or reducing agent

Worked example

Ethanol to ethanal (2019); butan-2-ol with CrO_3 (2017).

Use a **mild** oxidant: PCC (or anhydrous CrO_3) stops a primary alcohol at the aldehyde, $\text{CH}_3\text{CH}_2\text{OH} \xrightarrow{\text{PCC}} \text{CH}_3\text{CHO}$. Strong KMnO_4 or $\text{K}_2\text{Cr}_2\text{O}_7$ would give ethanoic acid. Butan-2-ol is secondary, so CrO_3 gives the ketone **butanone**.

- Ethanol to ethanoic acid — answer only: acidified KMnO_4 (or acidified $\text{K}_2\text{Cr}_2\text{O}_7$)
- Cyclohexenol (a ring with a $\text{C}=\text{C}$ and an $-\text{OH}$) to the unsaturated ketone — answer only: PCC; it oxidises the alcohol without touching the double bond
- Butan-2-one to butan-2-ol — answer only: NaBH_4 or LiAlH_4 , or H_2/Ni

5. Williamson synthesis: pick the halide and the alkoxide

Worked example

2-Methyl-2-methoxypropane, $(\text{CH}_3)_3\text{C}-\text{O}-\text{CH}_3$ (Q10(b), year not stated).

Split at the oxygen. One side is tertiary, the other methyl. The tertiary group must be the alkoxide and the methyl the halide: $(\text{CH}_3)_3\text{C}-\text{ONa} + \text{CH}_3\text{Br} \longrightarrow (\text{CH}_3)_3\text{C}-\text{O}-\text{CH}_3 + \text{NaBr}$ ($\text{S}_\text{N}2$). The other way round would give an alkene.

- 1-Propoxypropane — answer only: $\text{CH}_3\text{CH}_2\text{CH}_2\text{ONa} + \text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$
- Ethoxybenzene — answer only: sodium phenoxide + bromoethane ($\text{C}_6\text{H}_5\text{ONa} + \text{C}_2\text{H}_5\text{Br}$); not bromobenzene + sodium ethoxide
- 1-Methoxyethane — answer only: $\text{CH}_3\text{ONa} + \text{CH}_3\text{CH}_2\text{Br}$ (or $\text{CH}_3\text{CH}_2\text{ONa} + \text{CH}_3\text{Br}$)

6. Predict the products of an ether with HI

Worked example

$\text{CH}_3-\text{O}-\text{C}(\text{CH}_3)_3 + \text{HI}$ (2025).

A tertiary group is present: $\text{S}_\text{N}1$, a tertiary carbocation forms, iodide attacks it. Products: $(\text{CH}_3)_3\text{C}-\text{I} + \text{CH}_3\text{OH}$.

- Anisole with HI — answer only: phenol + CH_3I
- Ethoxyethane with excess HI — answer only: $2 \text{C}_2\text{H}_5\text{I} + \text{H}_2\text{O}$

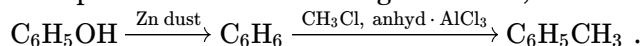
- $\text{CH}_3\text{--O--CH}_2\text{CH}_3 + \text{HI}$ — answer only: $\text{CH}_3\text{I} + \text{CH}_3\text{CH}_2\text{OH}$ (iodide on the smaller group)

7. Convert, using the reagent table

Worked example

Phenol to toluene (2019).

Heat phenol with zinc dust to get benzene; then Friedel–Crafts alkylation:



- Phenol to picric acid — answer only: conc. HNO_3 (poor yield), or conc. H_2SO_4 first and then conc. HNO_3 (good yield): 2,4,6-trinitrophenol
- Phenol to benzoquinone — answer only: $\text{Na}_2\text{Cr}_2\text{O}_7 / \text{H}_2\text{SO}_4$ (chromic acid)
- Chlorobenzene to phenol — answer only: fuse with NaOH at 623 K and 320 atm to $\text{C}_6\text{H}_5\text{ONa}$, then acidify
- Aniline to phenol — answer only: $\text{NaNO}_2 + \text{HCl}$ at 273–278 K, then warm with water
- Propanone to 2-methylpropan-2-ol — answer only: CH_3MgBr , then H_3O^+ (a ketone gives a tertiary alcohol)
- Ethanol to propan-2-ol — answer only: PCC to ethanal, then CH_3MgBr and hydrolysis (an aldehyde gives a secondary alcohol)
- Ethylmagnesium chloride to propan-1-ol (NCERT 7.20(iii)) — answer only: react with methanal, then hydrolyse
- Phenol to anisole — answer only: NaOH to sodium phenoxide, then CH_3Br
- Propene to propan-1-ol and to propan-2-ol — answer only: propan-1-ol by hydroboration–oxidation (B_2H_6 , then $\text{H}_2\text{O}_2/\text{OH}^-$); propan-2-ol by acid-catalysed hydration

8. The reasons that keep coming back

- **Alcohol boils higher than hydrocarbon or ether of the same mass:** intermolecular hydrogen bonding through the O--H .
- **Phenol is more acidic than an alcohol:** the phenoxide ion is stabilised by resonance; alkoxide has none (and the sp^2 carbon withdraws electrons).
- **o-Nitrophenol is steam volatile, p-nitrophenol is not:** intramolecular against intermolecular hydrogen bonding.
- **Tertiary alcohols resist oxidation:** no hydrogen on the --OH carbon.
- **tert-Butyl halide with an alkoxide gives an alkene:** steric hindrance, so the alkoxide acts as a base (elimination).
- **Phenol is easily nitrated or brominated:** --OH pushes electrons into the ring at ortho and para.
- **Phenol does not protonate readily:** the oxygen lone pair is delocalised into the ring and is less available.
- **Anisole with HI gives phenol, not an aryl iodide:** the aryl–O bond has partial double-bond character, and an sp^2 carbon will not undergo nucleophilic substitution.
- **The C–O bond in phenol is shorter than in methanol:** partial double-bond character (resonance, sp^2 carbon).

Past year questions · question types — ranked by how often they turn up

One video feeds this section: the "25 PYQ" video for CBSE 2026-27, which covers Q1–Q25, with questions from 2016 to 2025. Question numbers are the video's own. Years are as spoken in the video; where a year is spoken once for a numbered question it is applied to all its parts, and where no year was spoken the question is cited by number only. There is no second PYQ video for this chapter. Counts are by question part: 61 parts in all.

1 · Predict the product / complete the reaction — 2 or 3 marks · about 27 parts

Recognise it: "write the product of ...", "write the structure of the major product", "complete the following", "what happens when ...", "write the equation for ...".

1. Find the functional group that reacts and name the reaction type: reduction, oxidation, hydration, dehydration, ether cleavage, Grignard addition, electrophilic substitution, acetylation.
2. Match the reagent to the class (primary / secondary / tertiary; alkyl or aryl).
3. Write the organic product, plus the small by-product (H_2O , HCl , NaBr) if the question wants an equation.
4. For ring substitution give **both** ortho and para products and mark the para as major.

Trap: the reagent *selects* the product. NaBH_4 reduces a ketone and leaves an ester; PCC stops at an aldehyde and leaves a $\text{C}=\text{C}$; HI on an alkyl aryl ether gives phenol and alkyl iodide (never an aryl iodide); anisole needs no FeBr_3 ; copper at 573 K dehydrates a tertiary alcohol instead of dehydrogenating it.

2 · Give a reason / account for — 2 or 3 marks · about 12 parts

Recognise it: "give reason", "why is ...", "explain why ...", "account for ...".

1. Name the structural feature: hydrogen bonding (inter- or intra-molecular), resonance, partial double-bond character, steric hindrance, an electron-releasing or electron-withdrawing group, a missing hydrogen.
2. Say how that feature causes the behaviour, in one sentence, and compare with the other compound.

Trap: the answer is a mechanism-level sentence, not a restatement. "Phenol is more acidic" must go on to "because phenoxide is stabilised by resonance and alkoxide is not". "o-Nitrophenol is steam volatile" must name **intramolecular** against **intermolecular** hydrogen bonding.

3 · Convert A to B — 2 to 5 marks · about 10 parts

Recognise it: "how will you convert ...", "carry out the following conversion".

1. Say what changes: a functional group (alcohol to aldehyde), a carbon count (ethanol to propan-2-ol: one more carbon), or the class (primary to secondary).
2. Pick the named route: Grignard for adding carbon, a mild oxidant or reductant for changing oxidation level, diazonium for aniline, fusion with NaOH for chlorobenzene.
3. Write each step with its reagent over the arrow.

Trap: a secondary alcohol comes from an aldehyde (plus Grignard) or a ketone (plus a reducing agent); a tertiary alcohol comes from a ketone plus Grignard. Count the carbon neighbours of the product's $-\text{OH}$ carbon, then work backwards.

4 · Write the mechanism — 2 to 5 marks · 4 parts

Recognise it: "write the mechanism of ...", "show the curved-arrow notation ...".

1. Name the reaction and write the steps in order: protonation, nucleophilic attack or carbocation formation, loss of a proton.
2. Show the movement of electron pairs from the electron-rich atom to the electron-poor one, and the charges that result. Say which step is slow.
3. End by saying the acid is regained, so it is a catalyst.

Trap: hydration of an alkene (ethene plus water, H^+) is **three steps** and the first makes H_3O^+ ; ether formation at 413 K is an **$\text{S}_{\text{N}}2$** step (a second alcohol attacks the protonated one); alkene formation at 443 K goes through a **carbocation**, and that step is the slowest.

5 · Distinguish between two compounds — 2 or 3 marks · 3 parts

Recognise it: "give one chemical test to distinguish between ...".

1. Phenol against alcohol: neutral FeCl_3 , violet colour for phenol.
2. Primary against tertiary alcohol: Lucas reagent, turbidity at once for tertiary.
3. Ethanol against an ether: iodoform test, yellow precipitate for ethanol.

Trap: state the reagent, the observation for **each** compound (one shows, one does not), and not just "gives a colour".

6 · Arrange in order — 1 to 3 marks · 2 parts

Recognise it: "arrange in increasing order of acidic strength" and "arrange in increasing order of boiling point".

1. Acidity: electron-withdrawing group raises it, electron-releasing group lowers it, an alcohol is below any phenol.
2. Boiling point: hydrogen bonding > strong dipole (aldehyde) > weak dipole (ether) > non-polar alkane, and compare carbon counts last.

Trap: read "increasing" and write the order from weakest or lowest to strongest or highest.

7 · Write the IUPAC name — 1 mark · 2 parts

Recognise it: a drawn alcohol or ether, with "write the IUPAC name".

1. Alcohol: longest chain containing the $-\text{OH}$ carbon, number from the $-\text{OH}$ end, replace "e" by "ol".
2. Ether: the longer chain is the parent; the other side is "...oxy". A phenyl group on a chain is a "phenyl" substituent.

Trap: in 3-phenylprop-2-en-1-ol the $-\text{OH}$ gets number 1; the ring is a substituent, so it is named "phenyl" and gets a locant.

8 · Identify A, B, C, D, E — 5 marks · 1 part

Recognise it: "an organic compound A of formula ... reacts with ... to give B ...; identify A to E and write the reactions".

1. Use the molecular formula and the first clue (reaction with sodium gives hydrogen: an alcohol) to fix A.
2. Then follow each reagent: each gives one unique class of product.

Trap: $\text{C}_2\text{H}_6\text{O}$ could be ethanol or methoxymethane, but only the alcohol reacts with sodium. Always use the "reacts with sodium" clue to decide which.

Past year questions · mark slots — *what each type is worth*

The video states its own marks: Q1–Q10 are 2-mark short answers, Q11–Q23 are 3-mark short answers, and Q24–Q25 are 5-mark long answers. A "(Qn)" with no year was given without one in the video.

Question type	2 marks	3 marks	5 marks
Predict the product / complete	2017 Q1(b), 2018 Q2(a)(b), 2017 Q7(a)(b), 2020 Q9(a)(b), Q10(b)	2016 Q11, 2020 Q12, 2025 Q14, 2020 Q15, 2024 Q20, 2024 Q23	—
Give a reason	2023 Q6(a), 2020 Q8	2023 Q13(a)(c), 2019 Q16, 2019 Q19(b), 2025 Q22	2019 Q25(c)
Convert A to B	2019 Q5(a)(b), Q10(a)	Q17, 2019 Q19(a)	2019 Q25(a)
Mechanism	2017 Q3(b), 2023 Q4	2017 Q18	2019 Q25(b)
Distinguish by a test	2023 Q6(b)	2019 Q21(a)	—
Arrange in order	2017 Q3(a)	2023 Q13(b)	—
IUPAC name	2017 Q1(a)	2019 Q21(b)	—
Identify A–E	—	—	2016 Q24

Past year questions · repeat offenders — *appeared more than once — highest probability in the chapter*

Q1(b) · 2020 Q12(c) · 2023 Q13(c) · 2025 Q14(c)

Heating an ether with HI. The decision tree: primary and secondary on both sides, iodide goes on the smaller group ($\text{S}_{\text{N}}2$); one tertiary group, tertiary iodide ($\text{S}_{\text{N}}1$); alkyl aryl ether, phenol plus alkyl iodide because the aryl–O bond has partial double-bond character.

2018 Q2(b) · 2017 Q3(b) · 2017 Q18 · 2024 Q20(b)

Acid-catalysed hydration of an alkene, including the three-step mechanism for ethene to ethanol (asked twice in 2017) and the Markovnikov product for styrene and but-1-ene.

2020 Q15(iii) · Q17(ii) · 2024 Q20(a) · 2024 Q23(a)

A Grignard reagent with a ketone or an aldehyde, then hydrolysis: methylmagnesium bromide with propanone (tertiary alcohol), with ethanal (secondary), with cyclohexanone; cyclohexylmagnesium bromide with ethanal.

2019 Q5(b) · 2017 Q7(a) · 2020 Q9(b) · 2020 Q15(i) · 2024 Q23(d) · 2025 Q22(a)

Oxidation and dehydrogenation of alcohols: PCC to an aldehyde, CrO_3 to a ketone, copper at 573 K, and why tertiary alcohols resist oxidation.

2017 Q3(a) · 2019 Q16(c) · 2023 Q6(a)

Why phenol is more acidic than an alcohol, and the order for p-cresol, phenol, p-nitrophenol: resonance-stabilised phenoxide ion against unstabilised alkoxide.

Q10(a) · 2019 Q19(a) · 2023 Q13(c) · 2025 Q14(a)

Preparing phenol: from aniline (diazonium salt, warm water), from chlorobenzene (fuse with NaOH at 623 K, 320 atm), from anisole (cleave with HI) and from cumene (the industrial route, with acetone).

2023 Q4 · 2019 Q25(b)

The two mechanisms of acid dehydration of ethanol: 413 K gives ethoxyethane through S_N2; 443 K gives ethene through a carbocation.

2016 Q11(c) · 2024 Q23(c) · 2025 Q14(b)

Electrophilic substitution of anisole: Friedel–Crafts acylation and alkylation, and nitration; para major, ortho minor.

2016 Q11(a)(b) · 2019 Q25(c) · 2024 Q20(c) · 2024 Q23(b) · Q17(i)

Electrophilic substitution of phenol: Reimer–Tiemann, Kolbe, bromine water (white precipitate of 2,4,6-tribromophenol), bromine in CS₂ (mono), and nitration to picric acid; and why phenol reacts more readily than benzene.

Q10(b) · 2019 Q16(a)

Williamson synthesis: the pair that works (primary halide with a bulky alkoxide) and why a tertiary halide gives an alkene.

2019 Q16(b) · 2019 Q19(b) · 2023 Q13(b)

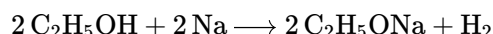
Hydrogen bonding as the explanation of boiling point and steam volatility: propanol against butane, o- against p-nitrophenol, the four-compound boiling-point order.

Past year questions · numerical types — *no calculations in this chapter, so these are the reaction and structure types — one worked model each, then cold practice*

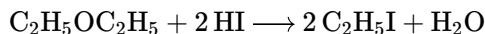
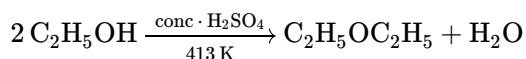
A · Identify A to E in a sequence

Worked example · 2016 Q24 — compound A, C₂H₆O, with sodium gives B and hydrogen; with I₂/NaOH gives C; with conc. H₂SO₄ at 413 K gives D, C₄H₁₀O; D with excess HI gives E

1. A reacts with sodium and releases hydrogen, so it has an O–H: an alcohol, not the isomeric ether methoxymethane. C₂H₆O as an alcohol is **ethanol** = **A**. B = sodium ethoxide.
2. The iodoform test (iodine and sodium hydroxide) needs a CH₃CH(OH)– group, which ethanol has: **C** = **iodoform** CHI₃, a yellow precipitate, with HCOONa as by-product.
3. Conc. H₂SO₄ at **413 K** is the temperature for the **ether** (443 K would give the alkene): two C₂H₅OH minus H₂O gives C₄H₁₀O = **D** = ethoxyethane.
4. Excess HI cleaves both ethyl groups: **E** = **ethyl iodide**.



plus 5 NaI + 5 H₂O.

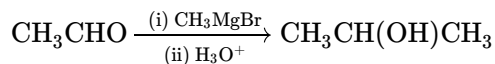
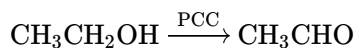


- 2016 Q24, cold — answer only: A ethanol; B sodium ethoxide; C iodoform (plus sodium methanoate); D ethoxyethane; E ethyl iodide

B • Convert A to B

Worked example • 2019 Q25(a)(ii) — ethanol to propan-2-ol

1. Propan-2-ol is a **secondary** alcohol with one more carbon than ethanol. A secondary alcohol is made from an aldehyde plus a Grignard reagent, so the route is: ethanol to ethanal, then add one carbon as CH₃—.
2. Oxidise ethanol with a **mild** oxidant (PCC or CrO₃) to ethanal.
3. Add CH₃MgBr to ethanal, then hydrolyse.



- 2019 Q5(a) — phenol to toluene — answer only: heat with zinc dust to benzene, then CH₃Cl with anhydrous AlCl₃ (Friedel–Crafts)
- 2019 Q5(b) — ethanol to ethanal — answer only: PCC (a mild oxidising agent)
- Q10(a) — aniline to phenol — answer only: NaNO₂/HCl at 0–5 °C, then warm with water
- Q17(i) — phenol to picric acid — answer only: concentrated HNO₃, giving 2,4,6-trinitrophenol
- Q17(ii) — propanone to 2-methylpropan-2-ol — answer only: CH₃MgBr then hydrolysis (a ketone gives a tertiary alcohol)
- Q17(iii) — butan-2-one to butan-2-ol — answer only: LiAlH₄ or NaBH₄, or H₂/Ni
- 2019 Q19(a) — phenol to benzoquinone, and chlorobenzene to phenol — answer only: Na₂Cr₂O₇/H₂SO₄; and fuse with NaOH at 623 K and then acidify
- 2019 Q25(a)(i) — phenol to anisole — answer only: NaOH to sodium phenoxide, then CH₃Br (Williamson)

C • Predict the product with a named reagent

Worked example • 2020 Q15 — three reactions

1. A cyclic alcohol with a C=C, treated with **PCC**: PCC oxidises the secondary alcohol to a ketone and leaves the double bond alone: an unsaturated ketone (cyclohexenone).
2. Salicylic acid with acetic anhydride and acid: the phenolic –OH is acetylated, giving **acetylsalicylic acid, aspirin** (a medicine).
3. Cyclohexanone with CH₃MgBr then hydrolysis: the methyl carbon attacks the carbonyl carbon and, after hydrolysis, gives the tertiary alcohol **1-methylcyclohexanol**, with Mg(OH)Br as by-product.

- 2018 Q2(a) — a ketone with an ester side chain, with NaBH_4 — answer only: the ketone becomes a secondary alcohol; the ester is not reduced
- 2018 Q2(b) — styrene $\text{C}_6\text{H}_5\text{CH}=\text{CH}_2$ with water and H^+ — answer only: $\text{C}_6\text{H}_5\text{CH}(\text{OH})\text{CH}_3$ (Markovnikov; OH on the carbon carrying the ring)
- 2017 Q7(a)(b) — butan-2-ol with CrO_3 , and with SOCl_2 — answer only: butanone; 2-chlorobutane
- 2020 Q9(a) — a compound with a benzylic $-\text{CH}_2\text{OH}$ and a phenolic $-\text{OH}$, heated with HCl — answer only: the benzylic $-\text{OH}$ becomes $-\text{Cl}$; the phenolic $-\text{OH}$ stays (partial double bond in the phenol $\text{C}-\text{O}$)
- 2020 Q9(b) — tert-butyl alcohol heated with copper at 573 K — answer only: 2-methylpropene and water (dehydration)
- 2020 Q12(a)(b) — propene by hydroboration–oxidation; tert-butyl alcohol with 20% H_3PO_4 at 358 K — answer only: propan-1-ol; 2-methylpropene
- 2024 Q20 — ethanal with CH_3MgBr then hydrolysis; but-1-ene with dilute H_2SO_4 ; phenol with bromine water — answer only: propan-2-ol; butan-2-ol; 2,4,6-tribromophenol (white precipitate)
- 2024 Q23 — cyclohexylmagnesium bromide with ethanal; phenol with Br_2/CS_2 ; anisole with $\text{CH}_3\text{Cl}/\text{AlCl}_3$; propan-2-ol over copper at 573 K — answer only: 1-cyclohexylethanol; o- and p-bromophenol (para major); 2- and 4-methoxytoluene (para major); propanone
- 2016 Q11 — Reimer–Tiemann; Kolbe's reaction; Friedel–Crafts acylation of anisole — answer only: salicylaldehyde; salicylic acid; 4-methoxyacetophenone (major) and 2-methoxyacetophenone
- 2025 Q14(a)(b) — phenol from cumene; nitration of anisole — answer only: cumene, then O_2 (air), then dilute acid, gives phenol and acetone; 4-nitroanisole (major), 2-nitroanisole

D • Write the mechanism

Worked example • 2017 Q18 — hydration of ethene to ethanol

- 1. Protonation of the alkene to a carbocation by electrophilic attack of H_3O^+ .**
 $\text{H}_2\text{O} + \text{H}^+ \longrightarrow \text{H}_3\text{O}^+$. The positive oxygen pulls the bonding electrons, so H_3O^+ gives up H^+ (and becomes H_2O). The H^+ is attacked by the π electrons of $\text{C}=\text{C}$: a $\text{C}-\text{H}$ bond forms and the other carbon is left with a positive charge: $\text{CH}_2=\text{CH}_2 + \text{H}_3\text{O}^+ \longrightarrow \text{CH}_3-\text{CH}_2^+ + \text{H}_2\text{O}$.
 - 2. Nucleophilic attack of water on the carbocation.** A lone pair on a water oxygen bonds to the positive carbon; the oxygen is now positive: $\text{CH}_3\text{CH}_2^+ + \text{H}_2\text{O} \longrightarrow \text{CH}_3\text{CH}_2-\text{OH}_2^+$.
 - 3. Deprotonation to form the alcohol.** The positive oxygen pulls the $\text{O}-\text{H}$ electrons, a water molecule takes the H^+ , and $\text{CH}_3\text{CH}_2\text{OH}$ forms; H_3O^+ is regenerated, so it is a catalyst.
- 2017 Q3(b) — ethene with H_3O^+ giving CH_3CH_2^+ and water, curved arrows — answer only: the $\text{O}-\text{H}$ electrons go to oxygen and release water plus H^+ ; the π electrons attack the H^+ ; a carbocation forms
 - 2023 Q4 — ethoxyethane from ethanol at 413 K — answer only: protonation; $\text{S}_\text{N}2$ attack of a second ethanol on the protonated alcohol with water leaving; loss of a proton, regenerating H^+
 - 2019 Q25(b) — ethene from ethanol at 443 K — answer only: protonated alcohol (fast); carbocation (slow, rate-determining); loss of a β proton gives ethene and returns H^+

E • Arrange in order

Worked example • 2023 Q13(b) — increasing boiling point: ethoxyethane, butanal, butanol, butane

1. n-Butane: non-polar, weak van der Waals forces: lowest.
2. Ethoxyethane: weakly polar, weak dipole interaction.

3. Butanal: strongly polar carbonyl, strong dipole–dipole attraction.

4. Butan-1-ol: hydrogen bonds, the strongest: highest.

Answer: butane < ethoxyethane < butanal < butan-1-ol.

- 2017 Q3(a) — increasing acidic strength: p-cresol, p-nitrophenol, phenol — answer only: p-cresol < phenol < p-nitrophenol

F • Distinguish by a chemical test

Worked example • 2019 Q21(a)(i) — propan-1-ol and 2-methylpropan-2-ol

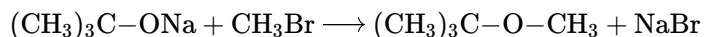
Add Lucas reagent (conc. HCl + anhydrous ZnCl₂) to each. The tertiary alcohol, 2-methylpropan-2-ol, forms the insoluble tertiary chloride at once and the solution turns cloudy. The primary alcohol, propan-1-ol, gives no turbidity at room temperature.

- 2019 Q21(a)(ii) — ethanol and dimethyl ether — answer only: iodoform test (I₂ + NaOH): ethanol gives a yellow precipitate of CHI₃; the ether gives none
- 2023 Q6(b) — phenol and cyclohexanol — answer only: neutral FeCl₃; phenol gives a violet complex, cyclohexanol does not react

G • Ethers: Williamson pairing and cleavage

Worked example • Q10(b) — 2-methyl-2-methoxypropane by Williamson synthesis

1. The ether is (CH₃)₃C–O–CH₃. Split at the oxygen: one side tertiary, the other methyl.
2. The alkyl halide must be **primary** (here methyl), so the tertiary group goes on the alkoxide.



The other pairing, (CH₃)₃C–Br + CH₃ONa, would give 2-methylpropene by elimination (2019 Q16(a)).

- Q1(b) — CH₃CH₂–O–CH(CH₃)₂ with HI — answer only: CH₃CH₂I + (CH₃)₂CHOH (iodide on the less hindered ethyl)
- 2020 Q12(c) — benzyl phenyl ether with HI (structure as drawn in the video; the spoken products are the benzyl iodide and phenol) — answer only: C₆H₅CH₂I + C₆H₅OH
- 2023 Q13(c) — phenol from anisole — answer only: heat with HI: CH₃I + phenol; the aryl–O bond has partial double-bond character so it does not break
- 2025 Q14(c) — methyl tert-butyl ether with HI — answer only: (CH₃)₃C–I + CH₃OH (S_N1, tertiary carbocation)
- 2019 Q16(a) — why tert-butyl bromide with sodium methoxide gives an alkene — answer only: three methyl groups block attack (steric hindrance), so methoxide acts as a base and removes a hydrogen: elimination to 2-methylpropene